

(12-13)

① $V_{rms} = 120V$
 $I_{rms} = \frac{V_{rms}}{Z} \cdot \cos \theta$
 $\frac{120}{945} = \frac{12006A}{945} \text{ or } 120.6mA$

② $P_{rms} = 120 \cdot 1206 \cdot 1005 = 1405$
 $P_{rms} = 120 \cdot 1206 \cdot 1005 = 1445W$

③ $f_c = 1.4MHz$ & $10kHz$
 $f_1 = 1.390$

$f_2 \pm 10kHz = f_1 \pm f_c$
 $f_2 = 1400MHz$
 $f_3 = 1410MHz$

Unit #5: Waves, Optics, & Medical Physics

① $B = \mu_0 \cdot I_d / 2\pi r$ $I_d = \Delta Q / \Delta t$
 $\Delta Q = 200nC = 2 \times 10^{-7}C$ $r = 1cm = 1 \times 10^{-2}m$
 $\Delta t = 2ns = 2 \times 10^{-9}s$

$\frac{4 \times 10^{-7}(1)}{2 \times 10^{-9}} = 2.0 \times 10^{-5} = 20\mu T$

② Displacement still generates a magnetic field
 $I_d = \Delta Q / \Delta t$

$\rightarrow 1.9$

(2-6)

① $D = C \cdot \Delta t$
 $C = 3 \times 10^8 m/s$
 $\Delta t = 10 \times 10^{-8} s$
 $D = 3000m$ round trip

② $R = \frac{(n_1 - n_2)^2}{(n_1 + n_2)^2}$
 $\frac{1 - 1.78^2}{1 + 1.78^2} = \frac{1.78^2}{2.78^2} = (0.2806)^2$

0.7873

③ $I = R/A$ $\frac{100W}{10 \times 10^{-4}m^2}$
 $I_c(r_1)^2 = 10 \times 10^4 (r_2)^2$
 $= 1 \times 10^8 (1/4)$
 $= 2.5 \times 10^7 W/m^2$

④ $n_1 = 1.50$, $n_2 = 1.0$, $n_3 = 1.0$
 $n_1 \sin \theta_1 = n_2 \sin \theta_2 = n_3 \sin \theta_3$
 $1.50 \sin 30^\circ = 1.0 \sin \theta_2$
 $\theta_2 = 45^\circ$
 $\theta_3 = 90^\circ$

⑤ $I = I_{inc} - I_{ref}$
 $I = 10W - 10W = 0W$
 $I = 10W$

⑥ $f = 3.247 \times 10^{15} s^{-1}$
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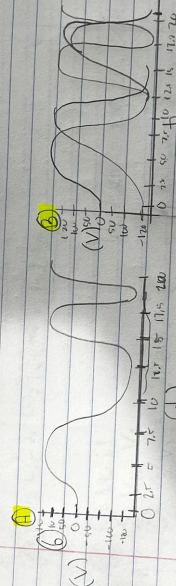
⑦ $RBE = 1$
 $RBE = 1$
 $RBE = 1$

(5-8)

⑤ $\frac{V_p}{V_s} = \frac{N_p}{N_s}$
 $\frac{240V}{120V} = \frac{N_p}{N_s}$

$N_p/N_s = 2$ (2:1)

$I_p/I_s = 1/2$ (1:2)



⑥ $V_p = 100V$ $V_s = 200V$
 $I_p = 2A$ $I_s = 1A$

⑦ $N_p = 200$ $N_s = 100$
 $I_p = 2A$ $I_s = 1A$

⑧ $P = VI$
 $P = 200V \times 1A = 200W$

⑨ $P = VI$
 $P = 200V \times 1A = 200W$

⑩ $P = VI$
 $P = 200V \times 1A = 200W$

(9-11)

① $20\mu s$
 $500ms$
 $1\mu s$
 $30ms$

$200ms/500ms$

$Y = L/R = L = R \cdot Y$

$(5 \times 10^5) 20 \times 10^{-9} = 10H$

② $R = L/Y \rightarrow 0.1/1 \times 10^{-9} = 1 \times 10^8 \Omega$

③ $100(20ms)$
 $100(10ms)$

④ $V_p/V_s = 1/2$

⑤ $V_p/V_s = 1/2$

⑥ $V_p/V_s = 1/2$

⑦ $V_p/V_s = 1/2$

⑧ $R = 0.1k\Omega = 1 \times 10^3 \Omega$

⑨ $C = 1\mu F = 1 \times 10^{-6} F$

⑩ $L = 10mH = 10 \times 10^{-3} H$

⑪ $f_0 = 1592.36Hz$

⑫ $f = 10Hz$ $10 > 1592.36Hz$

$\sqrt{10^4 + 983.04584}$

$\sqrt{991.045.84}$

$2 = 991.045.84$

$2 = 991.045.84$

$2 = 991.045.84$

(1-2)

Exam #2

Unit #4: Magnetism II

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A. force is left & velocity is up

(* Right hand thumb rule with index finger)
 $\vec{F}$ (left) & $\vec{v}$ (up) = $\vec{B}$ (into page)

B. force is into page & velocity is down
 $\vec{F}$ (into page) & $\vec{v}$ (down) = $\vec{B}$ (left)

C. force is up & velocity is left
 $\vec{F}$ (up) & $\vec{v}$ (left) = $\vec{B}$ (out of page)

A, B (into page)
B, B (left)
C, B (out of page)

(Right hand rule for loops & coil if flow clock wise)

MMF when closed for long time

- ① Coil = Counter Clock when closed, clockwise when open
- ② Coil = Counter clockwise when closed, clockwise when open
- ③ Coil = none X

(3-4)

③ A. $d = 2.20 \text{ cm} = .0220$
Avg 2.2 cm diameter
• 250s
• 4100 A

$$A = \pi r^2 = \pi \times (.0110)^2 = \pi \times .00121$$

$$B_F = 0T \quad \Delta B = .250$$

$$B_F = 2.00T \quad 2801 \times 10^{-4} \cdot B = 7.42 \times 10^{-4}$$

$$3.041 \times 10^{-3}, V = 3.041 \text{ mV (Avg emf)}$$

$$\text{B. } \mathcal{E} = 2.041 \times 10^{-3}V$$
$$R = .0100 \Omega \quad I = \frac{\mathcal{E}}{R} = \frac{2.041 \times 10^{-3}}{0.0100} = .3041 \text{ A}$$

$$\text{C. } P = I^2 R$$

$$I = .3041 \text{ A}$$
$$R = .0100 \Omega$$
$$P = (.3041)^2 \cdot .0100 = 9.25 \times 10^{-4} \text{ W} = .925 \text{ mW}$$

$$\text{E} = 18.0V \quad \text{④ } l = 10 \times 10^{-2} \quad W = 3.0 \times 10^{-4} \text{ m}^2$$
$$B = .640 \quad W = 3.0 \text{ cm} = 3.0 \times 10^{-2}$$
$$\text{Coil} =$$

$$l = 10 \text{ cm} = 18 / .640 \times (3.0 \times 10^{-2}) \cdot 18 \pi$$
$$W = 3.0 \text{ cm}$$

$$\text{Avg } V = 1875 \text{ mV} = 18 / .36$$
$$= 50$$

$$50 \text{ turns}$$

$$\text{⑤ } 2\pi / 18\pi$$

$$= .00335$$

$$I = .00335 \text{ A/cm}$$